LeadProductList Admin Module - Technical Documentation

1. Introduction

The LeadProductList Admin Module provides a web-based administrative interface for managing product configurations used in the Lead Creation system. It eliminates manual SQL operations through CRUD functionality, bulk Excel uploads, comprehensive validation, and full audit trail support.

2. Background & Problem Statement

The Lead Creation module fetches products from `MSTR_LD_CRTN_PRODUCT_TBL_VERSION_2` for lead generation. Prior to this module, product management required:

- Manual SQL script execution for all product additions/modifications
- Direct database access and privileges
- Coordinated database deployments for changes
- No systematic audit trail or change tracking

This created operational bottlenecks, increased error risk, and delayed product onboarding.

3. Solution Overview

Self-service web interface for product configuration management including:

- Single product add/edit via web forms
- Bulk Excel upload with template support
- Validation: duplicate detection, format checks, special character filtering
- Complete audit logging
- Role-based access control (Spring Security)
- Immutable Product IDs after creation

4. High-Level Architecture

Layered MVC architecture:

Controllers: `ProductListController` (CRUD), `ProductListUploadController` (Excel bulk upload) **Service:** `ProductListServiceImpl` - Business logic, validation, orchestration **DAO:** `ProductListDaoImpl` - Hibernate-based data access with upsert operations **View:** JSP with

`ProductListDaoImpl` - Hibernate-based data access with upsert operations **View:** JSP with JMesa tables, jQuery modals, client-side validation

Technology Stack: Spring 3.x (MVC, Security, Transactions), Hibernate 3.x, Apache POI, SQL Server, C3P0, JMesa

5. LeadProductList Module Functionality

5.1 Add Product

Single Product: Route `/admin/leadproductlist/saveAdd.htm`

- Input: Product ID, Name, Display Name, Active Status
- Validations: Product ID starts with "crm" (alphanumerics only), duplicate check on `iCRM_Product_ID`, special character filtering, UPPERCASE conversion
- Success: Insert with creator ID/timestamp | Failure: Error message with context

Bulk Excel Upload: Route `/admin/leadproductlist/ProductListUploadController.htm`

- Template: `Document\Lead_Product_List.xlsx` | Constraints: .xlsx, 0-2MB
- Workflow: Upload → Parse (POI XSSF) → Validate sheet/rows → Batch insert → Generate updated Excel with Success/Failure comments → Create audit record → Return file

5.2 Edit Product

Route `/admin/leadproductlist/viewEdit.htm?strPrimaryKey={productId}`, Save `/admin/leadproductlist/saveEdit.htm`

- **Product ID readonly** (immutable post-creation, commit bce4f930)
- Updatable: Name, Display Name, Active Status
- Validations: Same as Add | DAO upsert by `iCRM_Product_ID` | Auto-update modifier ID/timestamp

5.3 View/List Products

Route `/admin/leadproductlist/view.htm` - JMesa table with server-side pagination, filtering, sorting. Edit button opens modal.

6. Database Design Overview

Main Table: `MSTR_LD_CRTN_PRODUCT_TBL_VERSION_2`

Entity: `LeadProductListTbl.java`

Key Fields:

- `LD_PRD_ID` (PK, auto-increment), `iCRM_Product_ID` (Unique, VARCHAR 10, business key)
- `Product_Id`, `Product_Name`, `Product_Display_Name` (VARCHAR 100)
- `Business_Line_ID`, `Business_Line`, `Base_Product_Id`, `Base_Product_Name` (categorization/hierarchy)
- `Amount_Flag` (CHAR 1), `Is_Active` (CHAR 1: Y/N), `Layout_Key`, `category` (metadata)
- `Cre_Id`, `Cre_Time`, `Mod_Id`, `Mod_Time` (audit fields)

Defaults: `LEAD_AMOUNT_FLAG=N`, `LEAD_LAYOUT_KEY=NA`, `LEAD_CATEGORY=NA`

Audit Table: `Lead_Product_List_Audit_Tbl`

Entity: `LeadProductListAuditTbl.java` - Tracks bulk upload transactions

Fields: `id` (PK), `file_name` (VARCHAR 255), `create_date`, `mod_date`, `create_id` (VARCHAR 50), `success_count`, `failure_count`, `total_count` (INT), `upload_status` ("In Progress", "S", "F")

7. Data Flow

Admin to Database: User submits form/Excel → Controller → Service validates → DAO upserts (INSERT new with auto-ID / UPDATE existing by `iCRM_Product_ID`) → Transaction commits → Response

Main App Lead Creation: Query `MSTR_LD_CRTN_PRODUCT_TBL_VERSION_2` WHERE `Is_Active='Y'` → Populate product dropdown → Map selections to `iCRM_Product_ID`

8. Audit & Traceability

Record-Level: Each product tracks `Cre_Id` / `Cre_Time` (insert), `Mod_Id` / `Mod_Time` (edit), derived from Spring Security session

Transaction-Level: Bulk uploads log filename, timestamp, success/failure/total counts, upload status (In Progress → S/F), user accountability

Retrieval: `getLeadProductAuditByTime()` fetches audit records by timestamp for reconciliation

9. Security & Access Control

Authorization: Role `LD_PRD_LST` required (configured in `spring-security.xml`), enforced via Spring Security filter chain

Authentication: Form-based login, session fixation protection (migrateSession), user details via `FrameworkMstrUserGeneralDetails`

via [illegible]

Data Integrity: Product ID immutability, duplicate detection, special character filtering (injection mitigation)

10. Key Benefits

Operational Efficiency: Eliminates manual SQL, reduces onboarding from hours to minutes, enables non-technical users

Data Integrity: Comprehensive validation, duplicate detection, immutable Product IDs ensure referential integrity

Compliance: Complete audit trail, user accountability, bulk upload audit for compliance reporting

Scalability: Bulk upload for high-volume imports, pagination/filtering for large catalogs, transaction management for consistency

11. Conclusion

The LeadProductList Admin Module transforms product configuration management from manual SQL operations into a streamlined self-service interface. It demonstrates enterprise best practices: layered MVC architecture, comprehensive validation, audit logging, and role-based access control. This foundational component enables agile product management while maintaining data integrity and operational governance.

Branch: `73-i-refer-phase-4-product-list-menu` **Key Commits:**

- `bce4f930` - Product ID immutability enforcement
- `ec968081` - Enhanced validation for existing products and CRM ID format
- `e54ae063` - Special character validation implementation
- `8a787be5` - Edit functionality completion

Related Tables:

- `MSTR_LD_CRTN_PRODUCT_TBL_VERSION_2` (Main)
- `Lead_Product_List_Audit_Tbl` (Audit)

Access Role: `LD_PRD_LST`